SOC Home Lab Report: MITRE ATT&CK T1110.001 Brute Force Detection and Automated Response

1. Executive Summary

This report documents the design, implementation, and validation of an end-to-end brute-force detection and automated response workflow using Wazuh SIEM and native Windows Security Event telemetry.

The lab simulates a controlled password-guessing attack against a Windows 11 endpoint through SMB loopback authentication. Windows Security Event ID 4625 is collected by the Wazuh Agent and processed by the Wazuh Manager for detection and correlation. Individual failed authentication events are escalated into a composite brute-force detection when repeated failures against the same target account exceed the configured threshold within a defined time window.

The detection is mapped to MITRE ATT&CK Technique T1110.001 (Brute Force: Password Guessing). The simulation also validates the Windows Account Lockout mechanism through Event ID 4740 and demonstrates Wazuh Active Response as an additional containment layer. When the correlation rule is triggered, Wazuh automatically executes a Windows Firewall blocking action against the simulated source address, with the temporary containment rule automatically removed after the configured 180-second timeout.

The results demonstrate a complete security monitoring and response workflow consisting of event collection, detection, correlation, ATT&CK mapping, host-level defensive enforcement, automated containment, and response verification.

2. Threat Context & ATT&CK Mapping

The simulation models a password-guessing attack in which repeated authentication attempts are made against a local Windows account. The purpose of the exercise is to demonstrate how authentication failures can be collected, correlated, classified, and automatically contained using a SIEM-based detection and response workflow.

Parameter	Operational Detail
MITRE Tactic	Credential Access (TA0006)
MITRE Technique	Brute Force: Password Guessing (T1110.001)
Primary Data Source	Windows Security Event Log (Security.evtx), Event ID 4625
Enforcement Event	Account Lockout Event, Event ID 4740
Target Service	Server Message Block (SMB) / Windows network logon
Target Account	victimuser
Logon Type	Network Logon (Logon Type 3)

The simulation uses the loopback address 127.0.0.1 as the controlled source and target within the lab environment. This represents a local test scenario rather than a remote attacker simulation.

3. Attack Simulation Methodology

To simulate a controlled brute-force password-guessing attack, an automated PowerShell loop was executed against the local SMB service. The script issued 11 consecutive authentication requests against the victimuser account using intentionally invalid passwords.

The requests were generated at approximately 250-millisecond intervals to produce a sequence of failed authentication events within the correlation window configured in Wazuh.

Simulation Script:

```shell
1..11 | ForEach-Object {
    $null = net use \\127.0.0.1\c$ /user:victimuser "WrongPassword$_!" 2>&1
    Write-Host "Attempt $_ sent"
    Start-Sleep -Milliseconds 250
}
```

The simulation was performed in a controlled lab environment using a local test account and loopback SMB authentication. The objective was to generate repeatable authentication failure telemetry for detection and response validation.

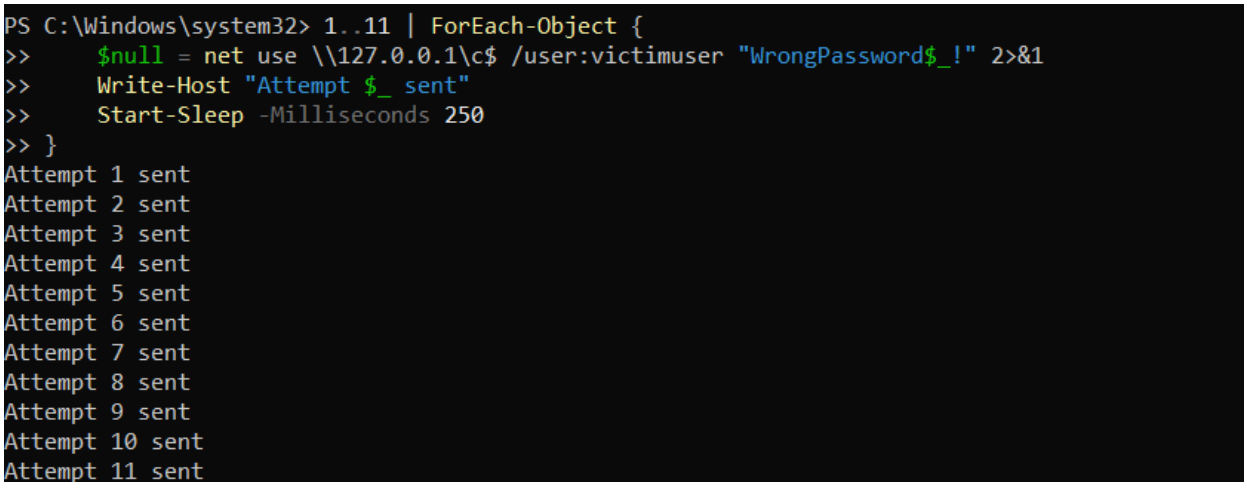


Figure 1: Administrator PowerShell console displaying execution of the 11-attempt brute-force password-guessing simulation.

4. Ingestion, Parsing & Detection Engineering Logic

The Wazuh Manager processes Windows Security Event telemetry forwarded by the Wazuh Agent. The detection workflow follows a three-stage escalation model:

4.1 Base Telemetry Ingestion — Rule 60122

- **Level:** 5
- **Description:** Logon Failure — Unknown user or bad password.
- **Event:** Windows Event ID 4625.
- **Condition:** Failed authentication with failure sub-status 0xC000006A, corresponding to an incorrect password.
- **Purpose:** Provides the atomic authentication-failure events used by the correlation rule.

4.2 Composite Correlation — Rule 60204

- **Level:** 10
- **Description:** Multiple Windows Logon Failures.
- **Correlation:** Multiple occurrences of Rule 60122 associated with the same target username.
- **Threshold:** 8 or more failed attempts.
- **Time Window:** 240 seconds.
- **Target:** victimuser.
- **MITRE Mapping:** T1110.001 (Brute Force: Password Guessing).
- **Purpose:** Escalates repeated authentication failures into a higher-confidence brute-force detection.

4.3 Defensive Policy Enforcement — Rule 60115

- **Level:** 9
- **Description:** User account locked out.
- **Event:** Windows Event ID 4740.
- **Purpose:** Confirms that the Windows account lockout mechanism responded to the repeated authentication failures.

This layered approach separates individual authentication telemetry from correlated detection and subsequent defensive enforcement.

5. Detection Telemetry & Evidence Verification

The detection workflow was verified using Wazuh alert telemetry and Windows Security Event data.
The evidence demonstrates the progression from individual failed authentication events to the composite brute-force detection and subsequent account lockout.

The verification focused on:

Successful ingestion of Windows Event ID 4625.
Detection of individual failed authentication events through Rule 60122.
Correlation of repeated failures through Rule 60204.
MITRE ATT&CK mapping associated with the correlated detection.
Generation of Windows Event ID 4740 following account lockout.
Execution of the configured Active Response mechanism.

	Oct 30, 2026 @ 19:01:47.446	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 19:01:47.445	Raakesh	User account locked out (multiple login e...	9	60115
	Oct 30, 2026 @ 19:01:45.042	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 19:01:42.648	Raakesh	Multiple Windows Logon Failures	10	60204
	Oct 30, 2026 @ 19:01:38.134	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 19:00:06.386	Raakesh	Windows Logon Success	3	60106
	Oct 30, 2026 @ 18:59:09.047	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 18:59:06.555	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 18:59:04.067	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 18:59:01.587	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 18:58:59.006	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122
	Oct 30, 2026 @ 18:58:54.391	Raakesh	Logon Failure - Unknown user or bad pa...	5	60122

Figure 2: Wazuh Threat Hunting Dashboard showing the chronological progression from individual logon failures to the composite brute-force alert and account lockout event.

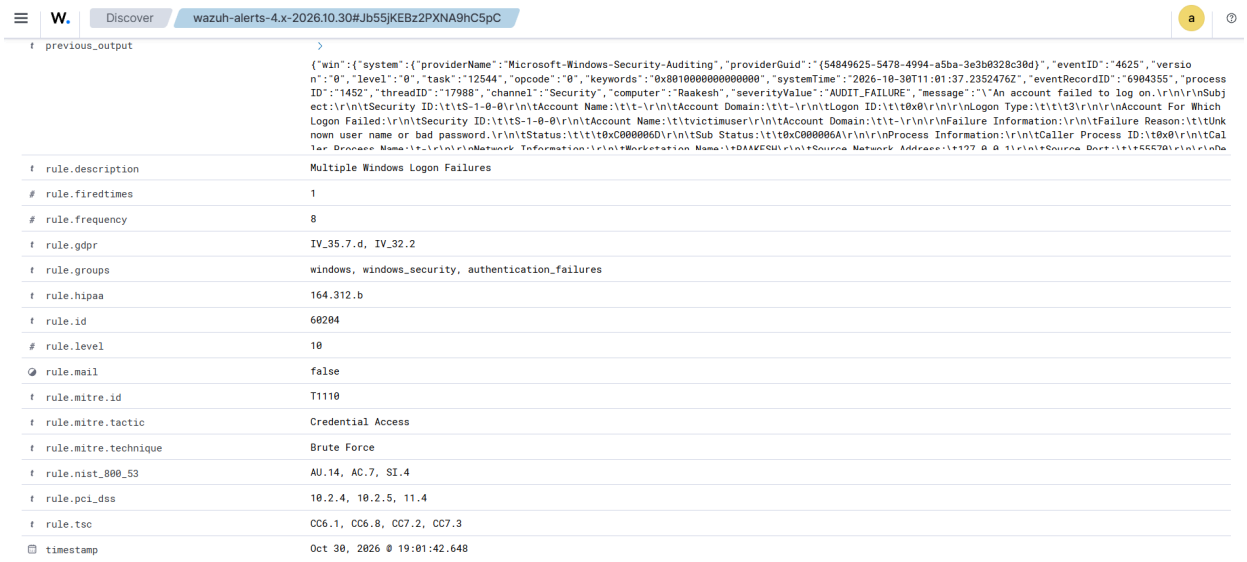


Figure 3: Expanded Wazuh alert payload for Rule 60204 showing the correlated Event ID 4625 metadata and associated MITRE ATT&CK mapping.

6. Engineering & Infrastructure Troubleshooting

During lab deployment and operation, several infrastructure issues were identified and resolved to maintain reliable SIEM operation.

6.1 Storage Exhaustion

The Wazuh Manager VM reached 100% disk utilization due to accumulated Vulnerability Detector CVE feed data, consuming approximately 13.7 GB of storage.
The issue was investigated by examining Wazuh storage usage and the Vulnerability Detector queue. The vulnerability-detection module was subsequently disabled in ossec.conf, and the accumulated data under /var/ossec/queue/vd was removed.
Following remediation, disk utilization was reduced from approximately 100% to 45%, restoring sufficient storage capacity for continued SIEM operation.

6.2 OpenSearch Write Protection

The previous disk exhaustion caused OpenSearch to enforce the read_only_allow_delete protection associated with the disk flood-stage condition. This prevented normal index write operations.
The OpenSearch cluster state was investigated and the write restriction was cleared using the appropriate administrative TLS client certificates. Normal index write operations were subsequently restored.

These troubleshooting activities demonstrated the importance of monitoring SIEM infrastructure health in addition to monitoring endpoint security events.

7. Incident Containment & Remediation

Host-Level Protection

The Windows Account Lockout Policy responded to the repeated failed authentication attempts by locking the targeted victimuser account. This prevented further authentication using the affected account until administrative remediation was performed.

Administrative Recovery

Following the controlled simulation, the test account was returned to an active state using the native Windows command:

```
Shell
net user victimuser /active:yes
```

The account status was subsequently verified via `net user victimuser` to confirm that it was no longer locked.

Automated Network Containment

In addition to the native Windows account lockout mechanism, Wazuh Active Response was configured to provide an additional network-level containment mechanism. When the composite Rule 60204 threshold was reached, Wazuh triggered a temporary Windows Defender Firewall block against the simulated source address.

Recommended Defensive Improvements

For a production environment, additional controls could include:

- Multi-Factor Authentication (MFA)
- Authentication rate limiting
- Stronger account lockout and authentication policies
- Monitoring of authentication failures across multiple endpoints
- Monitoring of Kerberos pre-authentication failures
- Centralized investigation and correlation of authentication activity

8. Automated Incident Containment — Wazuh Active Response

To demonstrate a closed-loop detection and response workflow, Wazuh Active Response was integrated with the brute-force correlation rule.
Rather than relying solely on passive alerting or Windows account lockout, the Active Response mechanism provided an additional automated containment layer at the endpoint firewall.

8.1 Active Response Configuration

Wazuh was configured to execute the native Windows netsh.exe utility when Rule 60204 was triggered on Agent 001.
The configuration consisted of:

- **Command:** netsh-drop
- **Executable:** netsh.exe
- **Location:** Defined agent
- **Agent:** 001
- **Trigger Rule:** 60204
- **Timeout:** 180 seconds

The response was bound specifically to Rule 60204 so that isolated failed authentication events would not independently trigger the network containment action.

8.2 Dynamic Containment

When the correlation threshold was reached, Wazuh dispatched the Active Response command to the Windows endpoint.
The response generated a temporary Windows Defender Firewall block targeting the simulated source address 127.0.0.1.
Because 127.0.0.1 was used as the loopback source for the controlled laboratory simulation, the result represents a local test of the response mechanism rather than a remote attacker isolation scenario.

8.3 Automated Cleanup

The Active Response was configured with a 180-second timeout. After the timeout expired, Wazuh automatically issued the corresponding cleanup operation to remove the temporary firewall block and restore the baseline firewall state.

8.4 Execution Verification

The Windows active-responses.log provided evidence of the complete response lifecycle:

1. **Detection:** Rule 60204 reached its configured threshold.
2. **Command Dispatch:** Wazuh Manager instructed the Windows agent to execute the Active Response.
3. **Dynamic Block:** The add command was executed at approximately 23:05:25 through active-response/bin/netsh.exe.
4. **Containment:** The Windows Defender Firewall applied the temporary block against the simulated source address.
5. **Automated Restoration:** The delete command was executed at approximately 23:08:26, approximately 180 seconds after the initial response.
6. **Teardown:** The temporary firewall block was removed.

```
2026/10/29 21:38:51 active-response/bin/restart-wazuh.exe: Starting
2026/10/29 21:38:51 active-response/bin/restart-wazuh.exe: Ended
2026/10/31 23:02:11 active-response/bin/restart-wazuh.exe: Starting
2026/10/31 23:05:14 active-response/bin/restart-wazuh.exe: Ended
2026/10/31 23:05:25 active-response/bin/netsh.exe: Starting
2026/10/31 23:05:25 active-response/bin/netsh.exe: {"version":1,"origin":{"name":"netsh.exe","module":"active-response"
},"command":"check_keys","parameters":{"keys":["127.0.0.1"]}}
2026/10/31 23:05:26 active-response/bin/netsh.exe: Ended
2026/10/31 23:08:26 active-response/bin/netsh.exe: Starting
2026/10/31 23:08:26 active-response/bin/netsh.exe: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"comm
and":"delete","parameters":{"extra_args":[],"alert":{"timestamp":"2026-10-31T15:05:24.124+0000","rule":{"level":10,"des
cription":"Multiple Windows Logon Failures","id":"60204","mitre":{"id":["T1110"],"tactic":["Credential
Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":1,"mail":false,"groups":["windows","windows_security",
"authentication_failures"],"pci_dss":["10.2.4","10.2.5","11.4"],"gdpr":["IV_35.7.d","IV_32.2"],"hipaa":["164.312.b"],"n
ist_800_53":["AU.14","AC.7","SI.4"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"001","name":"Raakesh","ip":
"192.168.0.136"},"manager":{"name":"wazuh-server"},"id":"1793459124.74663","previous_output":"{\"win\":{\"system\":{\"p
roviderName\":\"Microsoft-Windows-Security-Auditing\",\"providerGuid\":\"{54849625-5478-4994-a5ba-3e3b0328c30d}\",\"eve
```


Figure 4: Windows active-responses.log showing the Active Response add operation and subsequent delete operation after the configured timeout.

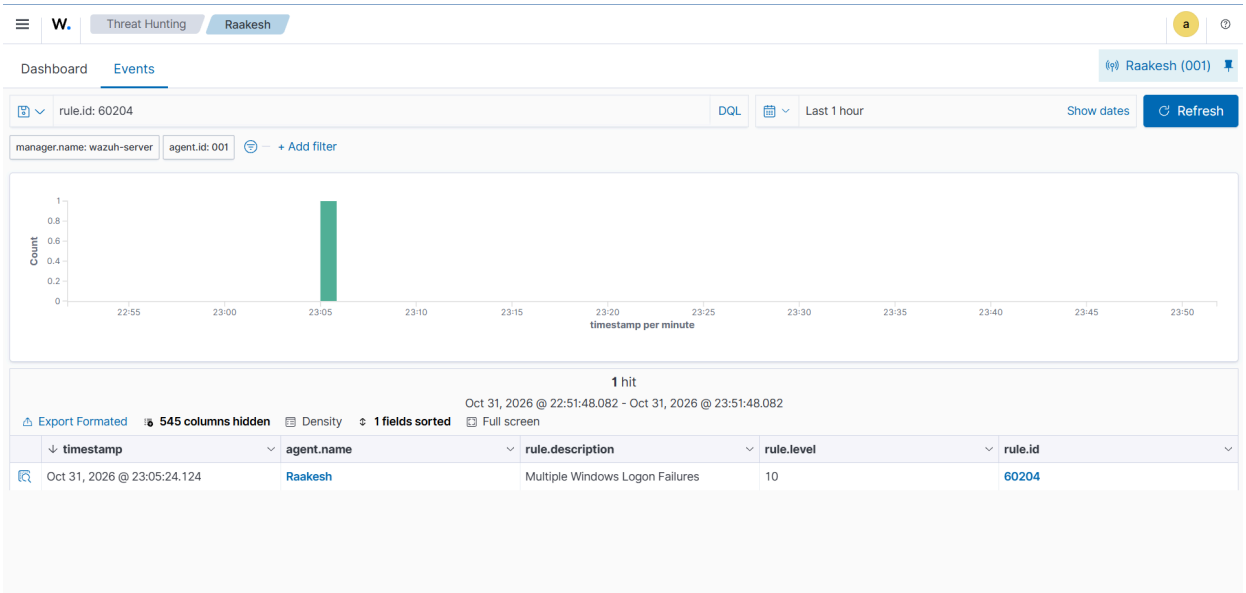


Figure 5: Wazuh Dashboard Active Response

9. Conclusion

This lab successfully demonstrated an end-to-end security monitoring and automated response workflow using Wazuh and Windows Security Event telemetry.

A controlled brute-force password-guessing simulation generated repeated Windows Event ID 4625 authentication failures. Wazuh successfully ingested and detected the individual events, correlated repeated failures using Rule 60204, and mapped the resulting detection to MITRE ATT&CK T1110.001.

The exercise also demonstrated the interaction between SIEM-based detection and host-level defensive mechanisms. Windows Account Lockout generated Event ID 4740, while Wazuh Active Response provided an additional automated firewall containment mechanism. The Active Response lifecycle was verified through both the add and delete operations recorded by the Windows endpoint.

The lab therefore demonstrates practical experience in security event monitoring, detection engineering, event correlation, MITRE ATT&CK mapping, SIEM troubleshooting, automated containment, and incident response verification.

10. Lab Limitations

The exercise was conducted in a controlled home-lab environment and therefore does not represent a production enterprise network.

The brute-force activity was simulated against the Windows endpoint using SMB loopback authentication through 127.0.0.1. Consequently, the Active Response firewall block was validated using a local test source rather than a remote attacker IP.

The exercise focused specifically on Windows authentication telemetry, Wazuh correlation, and automated response. It did not attempt to reproduce a complete enterprise identity infrastructure or a distributed multi-host attack.

These limitations define the scope of the experiment while providing a controlled environment for validating the detection and response workflow.